



ClasifEye: Face Detection and Gender & Age Classification

Visual Computing Project 2025-I

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Introduction & Justification

ClasifEye is an AI-driven system designed for face detection and the classification of gender and age. Our goal is to develop a tool with diverse applications while upholding ethical considerations.

This technology holds significant potential in personalized user experiences, demographic analysis for market research, and enhanced surveillance systems. A core focus is on analyzing and mitigating biases to ensure responsible and fair deployment.


```

def __init__(self, path):
    self.file = None
    self.fingerprints = set()
    self.logdups = True
    self.debug = debug
    self.logger = logging.getLogger(__name__)
    if path:
        self.file = open(os.path.join(path, 'fprints.txt'), 'a')
        self.file.seek(0)
        self.fingerprints.update(fprints)

    @classmethod
    def from_settings(cls, settings):
        debug = settings.getbool('debug')
        return cls(job_dir(settings), debug)

    def request_seen(self, request):
        fp = self.request_fingerprint(request)
        if fp in self.fingerprints:
            return True
        self.fingerprints.add(fp)
        if self.file:
            self.file.write(fp + os.linesep)

    def request_fingerprint(self, request):
        return request_fingerprint(request)

```

Project Objectives

Our primary objective was to develop an AI-based system capable of accurately detecting human faces and classifying gender and age range.

- **Technologies:** Python, OpenCV, TensorFlow/Keras for model development.
- **Model Architecture:** Leveraging ResNet50 with transfer learning for efficient and accurate feature extraction.
- **Dataset:** Utilizing the UTKFace dataset, specifically chosen for its rich annotations including age, gender, and race, crucial for training a diverse and robust model.

10-Week Project Plan

Our project followed a progressive, phased approach over 10 weeks to ensure thorough development and evaluation.

Weeks 1-2: Foundations

Research, environment setup, dataset (UTKFace) & tool selection.

1

Weeks 3-4: Preprocessing

Face detection (MTCNN/dlib), normalization



Weeks 5-6: Initial Model Training

Building & initial training of ResNet50 with Transfer Learning.

3

Weeks 7-8: Integration & Optimization

Integrating face detection, retraining, and fine-tuning model parameters.

4

Weeks 9-10: Evaluation & Docs

Final evaluation (accuracy, recall, F1-score), bias analysis, and comprehensive documentation.



Methodology: The ClasifEye Pipeline

Our system operates through a streamlined pipeline for efficient and accurate classification.

Core Pipeline:

1. **Face Detection:** Utilizing MTCNN/dlib for robust face localization.
2. **Preprocessing:** Normalizing and aligning detected faces for consistent input.
3. **Classification:** Feeding processed faces into the ResNet50 model for gender and age classification.

Preprocessing Details:

- **Image Normalization:** Resizing and scaling pixel values to a uniform range.
- **Age Normalization:** Applying MinMaxScaler to age values for better model convergence.
- **One-Hot Encoding:** Converting categorical features like gender and race into a numerical format suitable for model training.
- **Keypoint Use:** we used a model architecture focused on raw facial features and specific keypoints.



Pipeline Diagram: Input Image → Face Detection → Preprocessing → ResNet50 Classification → Age & Gender Output

UTKFace Dataset Exploration

Understanding our data was crucial for building a robust and fair AI model.



Data Extraction

Extracted age, gender, and race from filenames (e.g., "23_0_3...jpg") for structured analysis.



Key Visualizations

Analyzed data through age distribution histograms and gender/race counts.



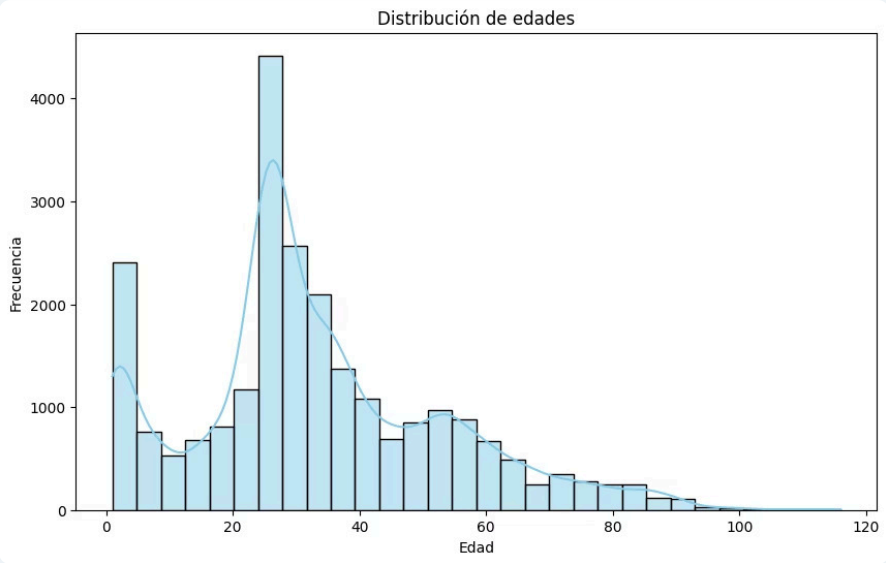
Critical Findings

Identified racial imbalance (fewer "Other" categories) and non-uniform age distribution.



Strategic Impact

These insights informed our data normalization and bias mitigation strategies.



Age Distribution in UTKFace Dataset



Data Preprocessing

Preparing Images and Labels for the Model

Preprocessed Face: Aligned and Normalized



Face Detection

Utilized **MTCNN/dlib** to accurately detect, crop, and align faces to a uniform 224x224 pixel resolution.



Normalization

Scaled image pixel values to **[0, 1]** and applied **MinMaxScaler** to age values for consistent model input.



Encoding

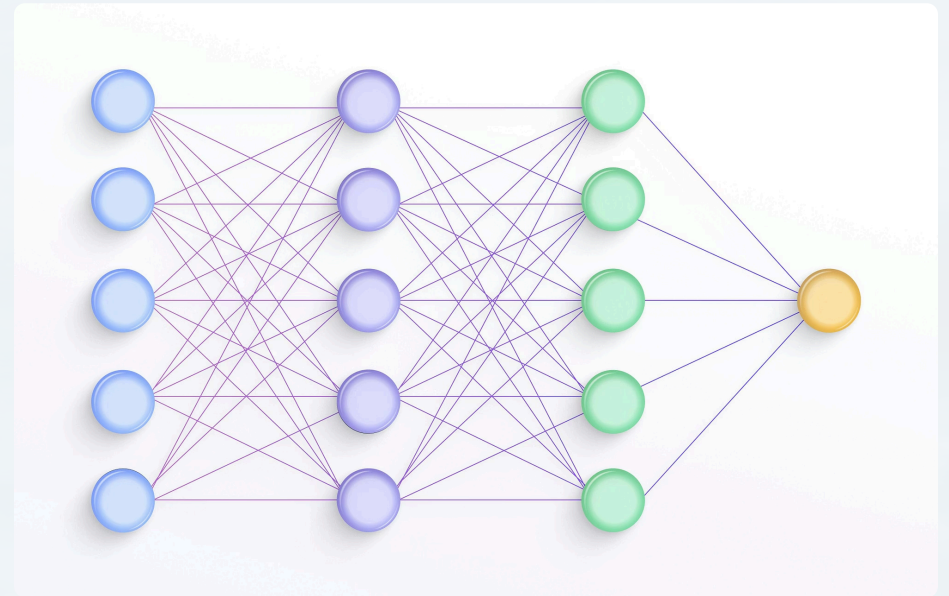
Converted categorical features like **gender and race** into numerical one-hot encoded vectors.

CNN Model: ResNet50

Gender, Race, and Age Classification

Model Architecture & Training

- **Architecture:** ResNet50 with Transfer Learning, pre-trained on ImageNet for robust feature extraction.
- **Outputs:** Designed for continuous age regression, and categorical classification for gender and race.
- **Inputs:** Uniformly processed 224x224 pixel images, optimized for model consumption.
- **Training:** Executed over 25 epochs using Adam optimizer, with MSE loss for age and categorical_crossentropy for gender/race.



Model Architecture Diagram



Technical Demo: ClasifEye in Action

We will now demonstrate ClasifEye's capabilities by processing images through our integrated pipeline.

The demonstration will showcase:

- **Image Loading:** How the system ingests new images for analysis.
- **Prediction Output:** Display of the classified gender and age range for each detected face.

Key Results & Metrics

Our model demonstrated strong performance across key metrics, validating the effectiveness of our approach.

~90%

Gender Accuracy

High precision in gender classification.

~85%

Race Accuracy

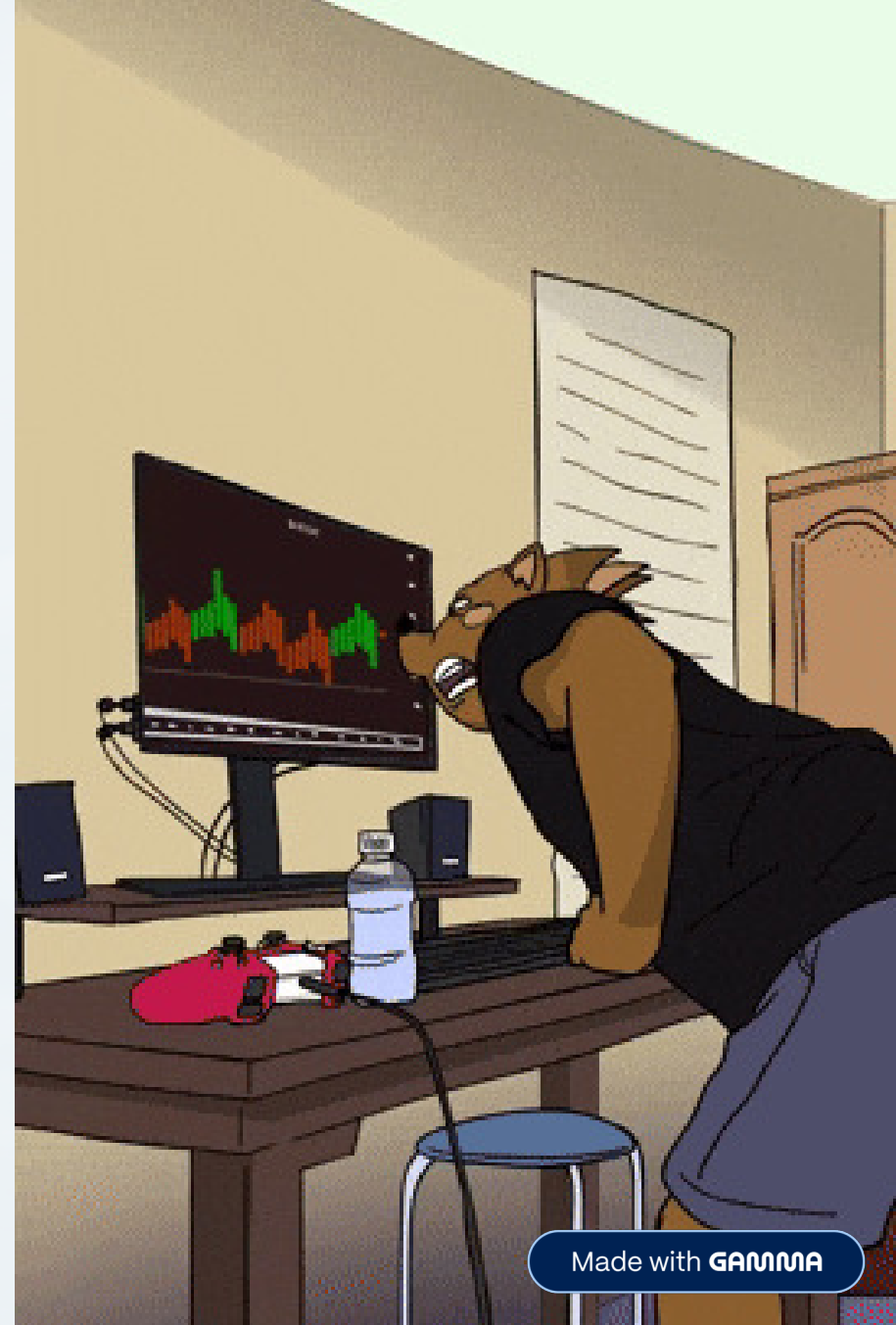
Robust classification across different racial categories.

~5-7...

Age MAE

Low Mean Absolute Error for age prediction, indicating high accuracy.

Bias Analysis: We observed some confusion in adjacent age ranges, a common challenge in age estimation. The end-to-end pipeline proved robust, handling real-world variations effectively.





HAPPY

Objectives Achieved!

We successfully met all SMART objectives within the 10-week timeframe, delivering a functional and effective system.

- **System Development:** A fully functional AI system capable of face detection and attribute classification was successfully built.
- **Progressive Approach:** Our phased methodology, from initial setup to iterative optimization, ensured continuous improvement and integration of components.
- **High Accuracy:** The system demonstrates high precision in gender and age classification, validating our model and dataset choices.
- **Iterative Refinement:** Continuous adjustments and retraining significantly enhanced overall performance, addressing challenges as they arose.

Learnings & Limitations

This project offered valuable insights into collaborative AI development, while also highlighting inherent limitations.

Key Learnings:

- **Collaborative Development:** Effective teamwork and version control were crucial for managing complex, interdependent tasks.
- **Library Mastery:** Deepened understanding and practical application of libraries like OpenCV, TensorFlow, and Keras.
- **Bias Analysis:** Gained critical experience in identifying and mitigating biases, promoting ethical AI practices.

Limitations:

- **Minority Class Confusion:** Potential for reduced accuracy in classifying less represented racial categories in the dataset.
- **Dataset Dependency:** Model performance is intrinsically linked to the quality and diversity of the UTKFace dataset, limiting its generalizability to highly unrepresented populations.



Ethical AI: Addressing biases for a fairer future.